Modbus_RTU Communication Specification

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Modbus_RTU Communication Specification Protocol numbers covered by this agreement Communication data format Frame format Device register addresses Fault code table Warning code description Power flow flag description CRC check algorithm

The agreement numbers included in this Agreement

- No.3, No.4, No.5, No.6, No.12

Communication data format

- During communication, data is transmitted in the form of words (WORD—2 bytes); within each transmitted word, the high byte precedes the low byte. If two consecutive words are transmitted (e.g., for a long integer), the high word precedes the low word.

Data type	Number of registers	Byte count	Explain
Character type	1	1	Send back two characters at a time; if fewer than two characters are available, use '0' to supplement.
Plastic surgery	1	2	In a single transmission, the high byte comes first, followed by the low byte.
Long rectangular type	2	4	Return the response in two characters: the higher character first, and the lower character last.

frame format

- ### Register Content Query (Function Code 03H)

  The start and end addresses of the query must correspond to the start and end addresses of a complete data block; otherwise, the returned data will be incorrect. For example, if the register start address for the device serial number is 186 and the register length is 12, the query start address must not be between 186 and 198; similarly, the end address (start address + number of registers to read) must also not fall within this range.

  Frame format sent by the host computer

Byte order	Code	Instance	Explain
0	Device address	01H	Device address (1–247)
1	03H	03H	FC
2	High byte of the start register address	00H	High 8 bits of the register address

Byte order	Code	Instance	Explain
3	Low byte of the start register address	10H	Low 8 bits of the register address
4	High byte of the number of registers	00H	The number of registers has an 8-bit high field.
5	Low byte of the number of registers	02H	Low 8 bits of the register count
6	High byte of CRC16 check	C0H	High 8 bits of CRC16 check
7	CRC16 low byte check	CBH	CRC16 check of the lower 8 bits

The slave machine has successfully parsed the format of the returned frame.

Byte order	Code	Explain
0	Device address	Device address (1–247)
1	03H	FC
2	Number of data bytes returned (N)	N = Number of requested registers × 2
3	The high byte of the first register's data	
4	First register: low byte of data	
.....	.....	
.....	.....	
	High byte of the data in register N	
	Low byte of the data in register N	
N+3	High byte of CRC16 check	All bytes prior to the CRC must be included in the checksum calculation.
N+4	CRC16 low byte check	

Lower-level device parsing of error feedback frame format

Byte order	Code	Explain
0	Device address	Device address (1–247)
1	03H	FC
2	Number of data bytes returned (N)	N = Number of requested registers × 2
3	The first 0	A total of N zeros are sent back.
4	The second zero	
.....	.....	
.....	.....	
	The (N-1)th zero	

Byte order	Code	Explain
	The N-th zero	
N+3	High byte of CRC16 check	All bytes prior to the CRC must be included in the checksum calculation.
N+4	CRC16 low byte check	

Register read example:

Read the data for the RMS value of the total mains current (starting register 202) and the total mains active power; specifically, the total mains current is reported as 5.0 A, the mains frequency as 50.0 Hz, and the total mains active power as 1200 W.

Host computer: 01 03 00 CA 00 03 25 F5

Slave device: 01 03 06 00 32 13 88 04 B0 9F 6B

- Register Content Setting (Function Code

  10H)Frame format sent by the host computer

Byte order	Code	Instance	Explain
0	Device address	01H	Device address (1–247)
1	10H	10H	FC
2	High byte of the start register address	01H	High 8 bits of the register address
3	Low byte of the start register address	10H	Low 8 bits of the register address
4	High byte of the number of registers	00H	The number of registers has 8 high bits (which are always equal to 0).
5	Low byte of the number of registers	02H	Low 8 bits of the register count
6	Number of bytes to be written (N)		N = number of registers × 2
7	The high byte of the first register's data		
8	The low byte of the first register's data		
.....	.....		
.....	.....		
	High byte of the data in register N		
	Low byte of the data in register N		
N+7	High byte of CRC16 check		High 8 bits of CRC16 check
N+8	CRC16 low byte check		CRC16 check of the lower 8 bits

The slave machine has successfully parsed the format of the returned frame.

Byte order	Code	Instance	Explain
0	Device address	01H	Device address (1–247)
1	10H	10H	FC
2	High byte of the start register address	01H	High 8 bits of the register address
3	Low byte of the start register address	10H	Low 8 bits of the register address
4	High byte of the number of registers	00H	The number of registers has 8 high bits (which are always equal to 0).
5	Low byte of the number of registers	02H	Low 8 bits of the register count
6	High byte of CRC16 check	41H	High 8 bits of CRC16 check
7	CRC16 low byte check	F1H	CRC16 check of the lower 8 bits

Lower-level device parsing of error feedback frame format

Byte order	Code	Explain
0	Device address	Device address (1–247)
1	90H	FC
2	Error code	Error code
3	High byte of CRC16 check	High 8 bits of CRC16 check
4	CRC16 low byte check	CRC16 check of the lower 8 bits

Error Code Description

Code	Explain
01H	Accessed a read-only register
03H	Writing data that exceeds the acceptable range
07H	Registers that cannot be modified under the current work mode

Register write example:

Set the output voltage (start register 606) to 220 V.

Host computer: 01 10 02 5E 00 01 02 08 98 8E 84

Slave device: 01 10 02 5E 00 01 61 A3

Device register address

- R: Indicates readable, i.e., supports the 03H command. W: Indicates writable, i.e., supports the 10H command.
- Int: Integer; Long: Long integer; UInt: unsigned integer; ULong: unsigned long integer; ASC: ASCII code
- Max: Return the maximum value; Min: Return the minimum value
- For the same address, the data returned varies depending on the protocol number; refer to the "Valid Protocol Number" column. For example, for an address 184, a device with protocol number 3 will return 3, while a device with protocol number 4 will return 4; if the valid protocol number field is empty, the data is considered invalid.
- The addresses in the table below are all expressed in decimal.

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Equipment fault code		ULong	100	2	R	There are 32 fault codes, each corresponding to one specific fault code; for details, see Faults. Code Table: Fault code 1 corresponds to bit 1, fault code 2 corresponds to bit 2, and so on.	3|4|5|6|12
Obtain the warning code			108	2		For details on the 32 warning codes, refer to the Warning Code Description.	3|4|5|6|12
Device type		UInt	171	1	R		3|4|5|6|12
Agreement Number		UInt	184	1	R	0x03	3
Agreement Number		UInt	184	1	R	0x04	4
Agreement Number		UInt	184	1	R	0x05	5
Agreement Number		UInt	184	1	R	0x06	6
Agreement Number		UInt	184	1	R	0x0C	12
Device serial number		ASC	186	12	R		3|4|5|6|12
Power flow status		UInt	198	1	R	For details, please refer to Power Bid Failure Position Description	3|4|5|6|12
Work pattern		UInt	201	1	R	0: Power-on mode 1: Standby mode 2: Grid Power Mode 3: Off-grid mode 4: Bypass Mode 5: Charging Mode 6: Failure Modes	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Total mains current RMS value	0.1A	Int	202	1	R		
Power frequency	0.01Hz	Int	203	1	R		3|4|5|6|12
Total mains active power	1w	Int	204	1	R		
Total TV power	1VA	Int	205	1	R		
Mains charging power	1w	Int	206	1	R		3|4|5|6|12
Power Flow State Extension		UInt	207	1	R	For details, please refer to Power Bid Failure Position Description	3|4|5|6|12
Reserved			208	18			
Total inverter current RMS value	0.1A	Int	226	1	R		
Inverter frequency	0.01Hz	Int	227	1	R		3|4|5|6|12
Total inverter active power	1w	Int	228	1	R		
Total inverter apparent power	1VA	Int	229	1	R		
Inverter charging current	0.1A	Int	230	1	R		3|4|5|6|12
Inverter temperature	1℃	Int	231	1	R		3|4|5|6|12
Reserved			232	20			
RMS total output current	0.1A	Int	252	1	R		4|6|12
Output frequency	0.01Hz	Int	253	1	R		3|4|5|6|12
Total active power output	1w	Int	254	1	R		4|6|12
Total output apparent power	1w	Int	255	1	R		4|6|12
Total load percentage	1%	Int	256	1	R		4|6|12
Reserved			257	20			

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Average battery voltage	0.1v	Int	277	1	R		3|4|5|6|12
Average filtered battery current	0.1A	Int	278	1	R	Positive values indicate charging, while negative values indicate discharging.	3|4|5|6|12
Average battery power	1w	Int	279	1	R		3|4|5|6|12
Battery percentage	1%	UInt	280	1	R		3|4|5|6|12
DCDC temperature	1℃	Int	281	1	R		3|4|5|6|12
Reserved			282	20			
Average total PV power	1w	Int	302	1	R		3|4|12
Average total charging power of PV systems	1w	Int	303	1	R		3|4|12
Average total charging current of PV	0.1A	Int	304	1	R		3|4|12
PV temperature	1℃	Int	305	1	R		3|4|5|6|12
Reserved			306	20			
Total external CT sampling power	1w	Long	326	2	R	Positive value: electrical energy output of the power grid; Negative values: Energy is being injected into the power grid.	
Reserved			328	10			
Mains voltage L1 RMS value	0.1v	Int	338	1	R		3|4|5|6|12
Mains voltage L1 RMS current	0.1A	Int	339	1	R		
Mains power L 1 active power	1w	Int	340	1	R		3|4|5|6|12
Mains power L1 apparent power	1VA	Int	341	1	R		
Inverter L1 RMS voltage	0.1v	Int	342	1	R		3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Inverter L1 RMS current	0.1A	Int	343	1	R		3|4|5|6|12
Inverter L1 active power	1w	Int	344	1	R	Positive values indicate the inverter output, while negative values indicate the inverter input.	3|4|5|6|12
Inverter L1 apparent power	1VA	Int	345	1	R		
Output L1 voltage RMS value	0.1v	Int	346	1	R		3|4|5|6|12
Output L1 RMS current	0.1A	Int	347	1	R		3|4|5|6|12
Output active power of L1	1w	Int	348	1	R		3|4|5|6|12
Output L1 apparent power	1VA	Int	349	1	R		3|4|5|6|12
Output L1 load percentage	1%	Int	350	1	R		3|4|5|6|12
PV1 average voltage	0.1v	Int	351	1	R		3|4|5|6|12
PV1 Average Current	0.1A	Int	352	1	R		3|4|5|6|12
Average PV1 power	1w	Int	353	1	R		3|4|5|6|12
L1 external CT sampling power	1w	Long	354	2	R	Positive value: electrical energy output of the power grid; Negative values: Energy is being injected into the power grid.	
Reserved			356	20			
Effective RMS value of mains voltage L2	0.1v	Int	376	1	R		
Mains voltage L2 RMS current	0.1A	Int	377	1	R		
Average power of the mains supply L2	1w	Int	378	1	R		

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Mains power L2 apparent power	1VA	Int	379	1	R		
Effective RMS voltage of the inverter L2	0.1v	Int	380	1	R		
Inverter L2 RMS current	0.1A	Int	381	1	R		
Average power of the inverter L2	1w	Int	382	1	R	Positive values indicate the inverter output, while negative values indicate the inverter input.	
Inverter L2 apparent power	1VA	Int	383	1	R		
Output L2 voltage RMS value	0.1v	Int	384	1	R		4|6|12
Output L2 RMS current	0.1A	Int	385	1	R		4|6|12
Output average L2 power	1w	Int	386	1	R		4|6|12
Output L2 apparent power	1w	Int	387	1	R		4|6|12
Output L2 load percentage	1%	Int	388	1	R		4|6|12
Average PV 2 voltage	0.1v	Int	389	1	R		3|4|12
Average PV2 current	0.1A	Int	390	1	R		3|4|12
Average PV2 power	1w	Int	391	1	R		3|4|12
L2 external CT sampling power	1w	Long	392	2	R	Positive value: electrical energy output of the power grid; Negative values: Energy is being injected into the power grid.	
Reserved			394	20			
Mains voltage effective value at L3	0.1v	Int	414	1	R		

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Mains voltage L3 RMS current	0.1A	Int	415	1	R		
Average L3 power of the mains supply	1w	Int	416	1	R		
Mains power L3 apparent power	1VA	Int	417	1	R		
Inverter L3 RMS voltage	0.1v	Int	418	1	R		
Inverter L3 RMS current	0.1A	Int	419	1	R		
Average power of the inverter L3	1w	Int	420	1	R	Positive values indicate the inverter output, while negative values indicate the inverter input.	
Inverter L3 apparent power	1VA	Int	421	1	R		
Output L3 voltage RMS value	0.1v	Int	422	1	R		
Output L3 RMS current	0.1A	Int	423	1	R		
Output average L3 power	1w	Int	424	1	R		
Output L3 apparent power	1w	Int	425	1	R		
Output L3 load percentage	1%	Int	426	1	R		
Average PV 3 voltage	0.1v	Int	427	1	R		12
Average PV3 current	0.1A	Int	428	1	R		12
Average PV3 power	1w	Int	429	1	R		12
L3 external CT sampling power	1w	Long	430	2	R	Positive value: electrical energy output of the power grid; Negative values: Energy is being injected into the power grid.	

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Reserved			432	168			
Output mode		Uint	600	1	R/W	0: Single-machine; 1: Parallel operation; 2: Three-phase combination-P1 3: Three-phase combination-P2 4: Three-phase combination-P3 5: Phase Separation-P1 6: Phase-separation-P2	3|4|5|6|12
(Primary) Output Priority		Uint	601	1	R/W	1: PV-Grid-Battery (SUB) [Priority Inverter] 2: PV Battery – Grid Power (SBU) 3: PV-Grid-Tied-Battery (SUF) [PV can be grid-tied]	3|4|5|6|12
(Secondary) Output Priority		Uint	602	1	R/W	0: Disable secondary output priority 1: PV-Grid-Battery (SUB) [Priority Inverter] 2: PV Battery – Grid Power (SBU) 3: PV-Grid-Tied-Battery (SUF) [PV can be grid-tied]	3|4|5|6|12
Start time of secondary output priority		Uint	603	1	R/W	Range: 0000–2359; the thousands and hundreds digits represent hours, while the tens and units digits represent minutes.	3|4|5|6|12
Secondary output priority end time		Uint	604	1	R/W	Range: 0000–2359; the thousands and hundreds digits represent hours, while the tens and units digits represent minutes.	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Output priority for the current time window		Uint	605	1	R	1: PV-Grid-Battery (SUB) [Priority Inverter] 2: PV Battery – Grid Power (SBU) 3: PV-Grid-Tied-Battery (SUF) [PV can be grid-tied]	3|4|5|6|12
Output voltage setpoint	0.1v	Uint	606	1	R/W	2200:220V output; 2300:230V output; 2400:240V output;	3|4|5|6|12
Output frequency setting value	0.01Hz	Uint	607	1	R/W	5000:50 Hz output; 6000:60 Hz output;	3|4|5|6|12
OP2 Overload Alarm Settings	1%	Uint	608	1	R/W	Range: 10% ~ 100%	4|6|12
Does OP2 allow output?		Uint	609	1	R/W	0: Prohibited 1: Permitted	4|6|12
OP2 output start time		Uint	610	1	R/W	Range: 0–23. If the start time is set to 0:00 and the end time is also set to 0:00, it indicates that OP2 can output whenever the conditions are met throughout the entire day.	4|6|12
OP2 output end time		Uint	611	1	R/W	Range: 0–23. If the start time is set to 0:00 and the end time is also set to 0:00, it indicates that OP2 can output whenever the conditions are met throughout the entire day.	4|6|12
Reserved			612	18			

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Battery type		Uint	630	1	R/W	0：AGM； 1：FLD； 2：USER； 4：Li2 6：Li4 8：LiB	3|4|5|6|12
Battery overvoltage protection point	0.1v	Uint	631	1	R/W		3|4|5|6|12
Battery charging priority		Uint	632	1	R/W	1: PV Priority (SOF) 2: PV system at the same level as the mains power supply (SNU) 3: PV charging only permitted (OSO) 4: PV shall prioritize satisfying the Output Remaining Charge (SOR) requirement.	3|4|5|6|12
Secondary charging priority		Uint	633	1	R/W	0: Disable secondary charging priority 1: PV Priority (SOF) 2: PV system at the same level as the mains power supply (SNU) 3: PV charging only (OSO) 4: PV shall prioritize satisfying the Output Remaining Charge (SOR) requirement.	3|4|5|6|12
Start time of secondary charging priority		Uint	634	1	R/W	Range: 0000–2359; the thousands and hundreds digits represent hours, while the tens and units digits represent minutes.	3|4|5|6|12
Secondary charging priority end time		Uint	635	1	R/W	Range: 0000–2359; the thousands and hundreds digits represent hours, while the tens and units digits represent minutes.	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Charging priority for the current period		Uint	636	1	R	1: PV Priority (SOF) 2: PV system at the same level as the mains power supply (SNU) 3: PV charging only permitted (OSO) 4: PV shall prioritize satisfying the Output Remaining Charge (SOR) requirement.	3|4|5|6|12
Maximum charging voltage	0.1v	Uint	637	1	R/W		3|4|5|6|12
Float charging voltage	0.1v	Uint	638	1	R/W		3|4|5|6|12
Constant voltage to float charge transition wait time	min	Uint	639	1	R/W	Range: 1–900 min; if set to 0, the default value is 10 min.	3|4|5|6|12
Maximum charging current	0.1A	Uint	640	1	R/W		3|4|5|6|12
Maximum mains charging current	0.1A	Uint	641	1	R/W		3|4|5|6|12
Maximum discharge current protection	1A	Uint	642	1	R/W	Range: 0–500 A	3|4|5|6|12
Mains power mode battery discharge recovery point [F]	0.1v	Uint	643	1	R/W	Scope: Max[12v*J , G+0. 1v] ~ (B-0.5v) Setting this to 0 indicates full recovery.	3|4|5|6|12
Mains power mode battery discharge protection point [G]	0.1v	Uint	644	1	R/W	Range: Max [11v*J,I+0.1 v] ~ Min[14.3 v*J,F-0.1v]	3|5
Mains power mode battery discharge protection point [G]	0.1v	Uint	644	1	R/W	Range: Max [11v*J,H+0.1 v] ~ Min[14.3 v*J,F-0.1v]	4|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Off-grid mode OP1 battery low-voltage protection point [H]	0.1v	Uint	645	1	R/W	Scope: I ~ Min[13.5v*J , G-0.1v]	4|6|12
Off-grid power Tank discharge protection point [I]	0.1v	Uint	646	1	R/W	Range: (10 V * J) ~ Min[G-0.1 V, 13.5 V *J]	3|5
Off-grid mode battery discharge protection point [I]	0.1v	Uint	646	1	R/W	Range: (10 V *J) ~ Min [H,13.5 V *J]	4|6|12
Mains power mode battery discharge SOC protection value [k]	1%	Uint	647	1	R/W	Range: Max[5%, N+1%] ~ Min[L-1% ,96%]	3|5
Protection value for battery discharge SOC in mains power mode [k]	1%	Uint	647	1	R/W	Range: Max[5%, M+ 1%] ~ Min[L-1%,96%]	4|6|12
Restoration value of battery discharge SOC in mains power mode [L]	1%	Uint	648	1	R/W	Range: Max [10%,K+ 1%] ~ Min[O, 100%]	3|4|5|6|12
Off-grid mode OP1: Battery discharge SOC protection value [M]	1%	Uint	649	1	R/W	Scope: N ~ Min[K-1%, 95%]	4|6|12
Off-grid mode battery discharge SOC protection value [N]	1%	Uint	650	1	R/W	Range: 0% ~ Min [K-1%,95%]	3|5
Off-grid mode battery discharge SOC protection value [N]	1%	Uint	650	1	R/W	Range: 0% ~ M	4|6|12
Battery Eq mode enabled		Uint	651	1	R/W	0: Prohibited; 1: Enable	3|4|5|6|12
Eqcharging voltage	0.1v	Uint	652	1	R/W	Scope: C ~ (A-1v)	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
bat_eq_time	min	Uint	653	1	R/W	Range: 0–900	3|4|5|6|12
Exit after Eq timeout	min	Uint	654	1	R/W	Range: 0–900	3|4|5|6|12
Interval between two Eq charges	day	Uint	655	1	R/W	Range: 1–90	3|4|5|6|12
Forced Eq charging		Uint	656	1	W	1: Manually force one charge cycle of Eq	3|4|5|6|12
Lithium-ion battery charging stop SOC [O]	1%	Uint	657	1	R/W	Upon successful communication with the lithium battery, retrieve the SOC value; stop charging once the battery reaches the preset SOC level. Range: Max[20%，L] ~100%	3|4|5|6|12
Maximum charge voltage of EEPROM	0.1v	Uint	658	1	R		3|4|5|6|12
EEPROM floating charge voltage	0.1v	Uint	659	1	R		3|4|5|6|12
Maximum charging current of EEPROM	0.1A	Uint	660	1	R		3|4|5|6|12
Reserved			661	16			
Input pattern		Uint	677	1	R/W	0：APL； 1：UPS； 2：GNT	3|4|5|6|12
Buzzer mode		Uint	678	1	R/W	0: Mute in all cases; 1: The input source triggers an alarm when subjected to a transformation, or when specific warnings or faults occur; 2: Sound when a specific warning or fault occurs; 3: Audible alert during failure	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
LCDbe in a poor light		Uint	679	1	R/W	0: Scheduled shutdown; 1: Chang Liang	3|4|5|6|12
LCD automatically returns to the home screen		Uint	680	1	R/W	0: Does not automatically return; 1: Automatically returns after 1 minute	3|4|5|6|12
Energy-saving mode switch		Uint	681	1	R/W	0: Energy-saving mode is turned off; 1: Energy-saving mode is enabled	3|4|5|6|12
Automatic restart on overload		Uint	682	1	R/W	0: Overload fault does not trigger a reboot; 1: Automatic restart on overload fault	3|4|5|6|12
Automatic restart upon overtemperature		Uint	683	1	R/W	0: Overtemperature fault does not trigger a restart; 1: Automatic restart on overtemperature fault	3|4|5|6|12
Overload bypass enable		Uint	684	1	R/W	0: Prohibited; 1: Enable	3|4|5|6|12
Parallel PV detection mode		Uint	685	1	R/W	0:; 1:	
External CT enabled		Uint	686	1	R/W	0: Prohibited; 1: Enable	
Warning Mask [I]		ULong	687	2	R/W	Warnings corresponding to 1 are displayed normally, while warnings corresponding to 0 are suppressed.	3|4|5|6|12
Dry contact		Uint	689	1	R/W	0: Normal mode; 1: Grounding box mode;	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Automatic mains power output enable		Uint	690	1	R/W	0: No mains power output unless the power button is pressed. 1: The device will automatically output power even if the power button is not pressed.	3|4|5|6|12
Maximum power output for PV grid-connected systems	1w	Uint	691	1	R/W	Range: 200 W to rated power	
Island detection enabled		Uint	692	1	R/W	0: Prohibited 1: Enable	
Power-on Method		Uint	693	1	R/W	0: Can be powered on locally or remotely; 1: Can only be booted locally; 2: Can only be powered on remotely	3|4|5|6|12
Remote switch		Uint	694	1	R/W	0: Remote shutdown 1: Remote Power-On	3|4|5|6|12
Exit fault lock state		Uint	695	1	W	1: Exit fault lock state (this action is only effective when the machine has entered fault mode)	3|4|5|6|12
Inverter date		Uint	696	3	R/W	Three 16-digit values representing year, month, and day, respectively (e.g., 07H E 6H 00H 07H) 00H 0EH indicates July 14,2022)	3|4|5|6|12
Inverter time		Uint	699	3	R/W	Three 16-bit values representing hour, minute, and second, respectively (e.g., 00H 0DH 00H 0FH 00H 10H represents 13:15:16).	3|4|5|6|12
Daily power generation	0.01kwh	Uint	702	1	R		3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
Gross generation	0.01kwh	ULong	703	2	R		3|4|5|6|12
Zero power generation		Uint	705	1	W	0xAA: Clear all power generation data	3|4|5|6|12
Reset user parameters			706	1	W	0xAA: User parameters are restored to their default values (takes effect when operating in non-off-grid mode)	3|4|5|6|12
Grounding relay enabled			707	1	R/W	0: Prohibited 1: Enable	4|6|12
Mains power identification time	s	Uint	708	1	R/W	The time required for the machine to progress from recognition to mains power detection and then to determination that the mains power is available. Range: 5–300	3|4|5|6|12
Lithium battery activation time	sec		709	1	R/W	Range: 6–300	3|4|5|6|12
MPPT multi-pulse power scanning mode		Uint	710	1	R/W	0: No scan (DIS) 1: Timed Scan (TIM) 2: Scan during power fluctuations (POW) 3: Both TIM and POW are active simultaneously (BTH)	3|4|5|6|12

Data name	Unit	Data format	Start address	Number of storage devices	Read-write	Explain	Effective Protocol Number
BMS Follow Mode		Uint	711	1	R/W	0: Neither the charging voltage nor the charging current follows the BMS values. 1: CV; Float voltage follows the BMS and is non-configurable; charging current does not follow the BMS. 2: CV, Float voltage, maximum charging current follows the BMS and is non-configurable 3: Charging stops based on the set SOC rather than voltage, and the charging current does not follow the BMS. 4: Charging stops based on the set SOC rather than voltage; the maximum charging current follows the BMS and is non-configurable.	3|4|5|6|12
Power rating	w	UInt	762	1	R		3|4|5|6|12
Rated number of battery cells [J]	PCS	UInt	763	1	R		3|4|5|6|12
Lithium battery automatic activation enabled		Uint	858	1	R/W	0: Manual activation; 1: Automatically activates once upon power-on.	3|4|5|6|12
Lithium battery activation: once		Uint	859	1	R/W	0: Invalid 1: Run one lithium battery activation cycle	3|4|5|6|12

Error Code Table

Fault code	Explain
bit0	Inverter overtemperature

Fault code	Explain
bit1	PV Overtemperature
bit2	DC-DC overtemperature
bit3	Battery overvoltage
bit4	Reserved
bit5	Output short circuit
bit6	Inverter overvoltage
bit7	Output overload
bit8	Busbar overvoltage
bit9	Busbar soft-start timeout
bit10	PVovercurrent
bit11	PVovervoltage
bit12	DCDCovercurrent
bit13	Inverter overcurrent
bit14	Busbar low voltage
bit15	Retain
bit16	The inverter's DC component is excessively high.
bit17	The PV2 current has excessive zero bias.
bit18	Excessive zero offset in the output current
bit19	Excessive zero offset of the inverter current
bit20	The zero offset of the battery current is too large.
bit21	PV1: excessive current offset
bit22	Inverter low voltage
bit23	Inverter reverse power protection
bit24	Host failure in parallel system
bit25	Commonly used synchronization signals for parallel systems
bit26	Battery type incompatibility
bit27	The parallel versions are incompatible.
bit28	Repeat external CT scan
bit29	Excessive zero offset in the output current 2

Warning Code Description

The system warning is a 32-bit unsigned long integer, where each bit corresponds to one warning; each bit can also be masked using the warning mask I. Once masked, the corresponding warning will no longer be readable on the LCD, nor will it be readable via commands.

- ## Warning Code Table

Warning Code	Explain
bit 0	Loss of zero-crossing in mains power
bit 1	Are you frequently observing the mains voltage waveform?
bit 2	Mains voltage overvoltage
bit 3	Mains voltage drop
bit 4	Mains power overfrequency
bit 5	Mains power frequency deficiency
bit 6	PV1 under-voltage
bit 7	Overtemperature
bit 8	The battery voltage is low.
bit 9	Battery not connected
bit 10	Output overload
bit 11	Battery equivalent charge
bit 12	The battery is at a low discharge voltage and has not yet reached its recharge recovery point.
bit 13	Output power derating
bit 14	Fan jamming
bit 15	The PV power is too low to be usable.
bit 16	Parallel communication failure
bit 17	Inconsistent single-parallel output mode
bit 18	Excessive voltage difference between parallel batteries
bit19	Lithium battery communications: your frequent choice
bit20	The battery discharge current has exceeded the set value.
bit 21	Island Detection
bit 22	PV2 undervoltage
bit23	Main output overload

Warning Code	Explain
bit24	The Sec output load exceeds the set value.
bit25	Sec output overload
bit 26~31	Retain

Explanation of Power Flow Indicators

The power flow is represented by a 16-bit unsigned integer, and the meaning of each bit is as follows:

- Electric current level gauge (corresponding to register 198)

Position	Explain
bit0~1	0: PV not connected to the system 1: PV has been connected to the system
bit2~3	0: Main power supply not connected to the system 1: Main power has been connected to the system.
bit4~5	0: Battery does not charge or discharge 1: Battery Charging 2: Battery discharge
bit6~7	0: The system does not output under load. 1: System output under load
bit8	0: Main power not charged 1: Grid power charging
bit9	0: PV not charged 1: PV charging
bit10	0: Battery icon is lit 1: Battery icon is off
bit11	0: PV icon lit up 1: PV icon is off
bit12	0: City traffic light indicator illuminated 1: The city map icon is disabled
bit13	0: Load icon lit 1: Load icon is off

- Power flow meter expansion (corresponding to register 207)

Position	Explain
bit0	0: PV1 not connected to the system; 1: PV1 connected to the system
bit1	0: PV2 not connected to the system; 1: PV2 connected to the system
bit2~3	0: AC Input 1 is not connected to the system 1: AC Input 1 has been connected to the system
bit4~5	0: AC input 2 is not connected to the system 1: AC input 2 has been connected to the system
bit6-7	0: AC input 3 is not connected to the system 1: AC input 3 has been connected to the system
bit8	0: Main output under load 1: The Main path does not output.
bit9	0: Sec-channel output under load 1: No output on Sec
bit10-15	Reserved

CRC校验 Algorithm

Parameter Model: CRC-16/MODBUS X16+X15+X2+1

C language code

```
const char auchCRCHi[] = {
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
```

```
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40
} ;

const char auchCRCLo[] = {
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06,
0x07, 0xC7, 0x05, 0xC5, 0xC4, 0x04, 0xCC, 0x0C, 0x0D, 0xCD,
0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09,
0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A,
0x1E, 0xDE, 0xDF, 0x1F, 0xDD, 0x1D, 0x1C, 0xDC, 0x14, 0xD4,
0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3,
0xF2, 0x32, 0x36, 0xF6, 0xF7, 0x37, 0xF5, 0x35, 0x34, 0xF4,
0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A,
0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29,
0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF, 0x2D, 0xED,
0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60,
0x61, 0xA1, 0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67,
0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F,
0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68,
0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA, 0xBE, 0x7E,
0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71,
0x70, 0xB0, 0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92,
0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C,
0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B,
0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89, 0x4B, 0x8B,
0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42,
0x43, 0x83, 0x41, 0x81, 0x80, 0x40
} ;

unsigned short sModbusCrc16(INT8U *chMsg, INT16U dataLen)
{
    unsigned char ubCRCHi = 0xFF;
    unsigned char ubCRCLo = 0xFF;
    unsigned char duwIndex;
    while (dataLen --)
    {
        duwIndex = 0xff&(ubCRCHi ^ *chMsg++);
        ubCRCHi = 0xff&(ubCRCLo ^ auchCRCHi[duwIndex]);
        ubCRCLo = auchCRCLo[duwIndex];
    }
    return (ubCRCHi << 8 | ubCRCLo);
}
```